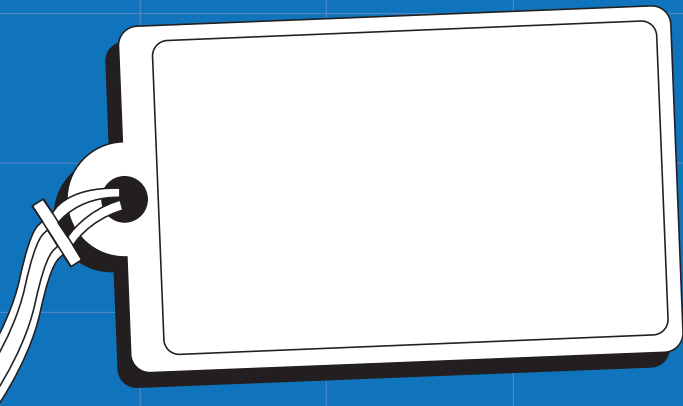




Interest

INTEREST

Solutions

Series **K**



MATHLETICS

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Page 5 questions

Find R as a decimal if:

a $r = 12\% \text{ pa}$

$$R = \frac{r}{100}$$

$$R = \frac{12}{100}$$

$$R = 0.12$$

b $r = 22\% \text{ pa}$

$$R = \frac{r}{100}$$

$$R = \frac{22}{100}$$

$$R = 0.22$$

Find the simple interest for the following:

a $P = \$1200, R = 0.08\%, T = 12$

$$I = PRT$$

$$I = \$1200 \times 0.08 \times 12$$

$$I = \$1152$$

b $P = \$2000, r = 12\%, T = 8$

$$I = P \times \frac{r}{100} \times T$$

$$I = \$2000 \times \frac{12}{100} \times 8$$

$$I = \$1920$$

Find the simple interest for the following:

a \$4550 at a rate of 20% pa for 4 years

$$P = \$4550, r = 20\%, T = 4$$

$$I = P \times \frac{r}{100} \times T$$

$$I = \$4550 \times \frac{20}{100} \times 4$$

$$I = \$3640$$

b \$32 000 for 2 years at a rate of 16% pa

$$P = \$32\,000, T = 2, r = 16\%$$

$$I = P \times \frac{r}{100} \times T$$

$$I = \$32\,000 \times \frac{16}{100} \times 2$$

$$I = \$10\,240$$

c \$50 000 for 3 months at a rate of 4.2% per month

$$P = \$50\,000, T = 3, r = 4.2\%$$

$$I = P \times \frac{r}{100} \times T$$

$$I = \$50\,000 \times \frac{4.2}{100} \times 3$$

$$I = \$6300$$

Page 6 questions

Convert the rate or the time periods to ensure they match

a $r = 3\% \text{ pa}, T = 12 \text{ months}$

$$T = 12 \text{ months} = 1 \text{ year}$$

Or

$$r = \frac{3}{12}\%$$

$$r = 0.25\% \text{ per month}$$

b $r = 7.8\% \text{ pa}, T = 156 \text{ weeks}$

$$T = 156 \text{ weeks} = \frac{156}{52} = 3 \text{ years}$$

Or

$$r = \frac{7.8}{52} = 0.15\% \text{ per week}$$

c $r = 9\% \text{ per month}, T = 2 \text{ years}$

$$T = 2 \text{ years} = 2 \times 12 = 24 \text{ months}$$

Or

$$r = 9 \times 12 = 108\% \text{ pa}$$

d $r = 0.05\% \text{ per week}, T = 2.08 \text{ years}$

$$T = 2.08 \times 52 = 108.16 \text{ weeks}$$

Or

$$r = 0.05 \times 52 = 2.6\% \text{ pa}$$

e $r = 1\% \text{ per day}, T = 1 \text{ years}$

$$T = 1 \times 365 = 365 \text{ days}$$

Or

$$r = 1 \times 365 = 365\% \text{ pa}$$

f $r = 6\% \text{ pa}, T = 1095 \text{ days}$

$$T = \frac{1095}{365} = 3 \text{ years}$$

Or

$$r = \frac{6}{365} = 0.016\% \text{ per day}$$

You invest \$4000 at a rate of 1.5% per month for three years.

a Find the simple interest

$$I = P \times \frac{r}{100} \times T$$

$$P = \$4000, T = 3 \text{ years}, r = 1.5\% \text{ per month}$$

Convert rate:

$$r = 1.5 \times 12 = 18\% \text{ pa}$$

$$I = \$4000 \times \frac{18}{100} \times 3$$

$$I = \$2160$$

b What is the total amount of money you now own?

$$\text{Total money} = \text{Principal} + \text{Interest}$$

$$\text{Total money} = \$4000 + \$2160$$

$$\text{Total money} = \$6160$$

$$\text{The total money you now own is } \$6160$$

Page 7 questions

A principal of \$7650 returns simple interest of \$3672 after 4 years.

a $I = \$3672, P = \$7650, T = 4$

$$I = PRT$$

$$3672 = 7650 \times R \times 4$$

$$R = \frac{3672}{7650 \times 4}$$

$$R = 0.12$$

$$R = \frac{r}{100}$$

$$0.12 = \frac{r}{100}$$

$$r = 12\% \text{ pa}$$

b Convert this interest rate to a monthly interest rate

$$r = 12\% \text{ pa}$$

$$r = \frac{12}{12}$$

$$r = 1\% \text{ per month}$$

Let's say you have \$5000 to invest for 5 years. Which of these rates returns the most interest?

- **2.7% per month**

change 2.7% per month to pa

$$r = 2.7 \times 12 = 32.4\% \text{ pa}$$

- **32% per month**

$$32\% \text{ pa}$$

- **0.7% per week**

change 0.7% per week to pa

$$r = 0.7 \times 52 = 36.4\% \text{ pa}$$

Since 0.7% per week gives the highest rate of return per annum it will be the rate that returns the most interest.

Page 8 questions

Your friend Jesse took out a loan of \$9840 to pay for a car. Jesse paid back the money by making \$250 payments each month for 4 years on a simple interest rate.

- a** What is the total amount Jesse paid back to the bank over the 4 years?

$$T = 4 \text{ years} = 4 \times 12 \text{ months} = 48 \text{ months}$$

$$\text{Payment } \$250 \times 48 = \$12\,000$$

- b** How much extra did Jesse pay on top of the loan?

$$\text{Extra paid} = \text{Total payment} - \text{loan}$$

$$\text{Extra paid} = \$12\,000 - \$9\,840$$

$$\text{Extra paid} = \$2\,160$$

- c** What was the monthly interest rate used to pay off the interest in part **b**?

$$I = PRT$$

$$\$2\,160 = \$9\,840 \times R \times 48$$

$$R = \frac{\$2\,160}{\$9\,840 \times 48}$$

$$R = 0.0046$$

$$R = \frac{r}{100}$$

$$0.0046 = \frac{r}{100}$$

$$r = 0.46\% \text{ per month}$$

- d** Convert this interest rate to an annual rate

$$r = 0.46 \times 12 = 5.5$$

$$r = 5.5\% \text{ pa}$$

Page 12 questions

A principal amount of \$4500 earns \$2100 interest over time

- a** Write down the values of I and P

$$I = \$2100$$

$$P = \$4500$$

- b** Find the value of the final amount A

$$A = I + P$$

$$A = \$2100 + \$4500$$

$$A = \$6600$$

\$6800 earns simple interest at a rate of 3.2% per month for 2 years.

- a** Use the simple interest formula to find the total value of this investment to the nearest cent?

$$I = \$6800, r = 3.2\%, T = 2 \text{ years} = 2 \times 12 \text{ months} = 24 \text{ months}$$

$$I = P \times \frac{r}{100} \times T$$

$$I = \$6800 \times \frac{3.2}{100} \times 24$$

$$I = \$5222.40$$

$$A = I + P$$

$$A = \$5222.40 + \$6800$$

$$A = \$12\,022.40$$

- b** What was the interest earned on this investment to the nearest cent?

$$I = \$5222.40$$

Page 13 questions

Find the value of the total investment (future value) and the interest I after one year to the nearest cent if

- a \$1200 is invested at a rate of 5% pa compounded annually

$$P = \$1200, r = 5\%, n = 1$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$1200\left(1 + \frac{5}{100}\right)^1$$

$$A = \$1200 \times 1.05$$

$$A = \$1260$$

$$A = P + I$$

$$\$1260 = \$1200 + I$$

$$I = \$1260 - \$1200$$

$$I = \$60$$

- b \$16 700 is invested at 12% pa compounded quarterly

$$P = 16\,700, r = \frac{12}{4} = 3\% \text{ quarterly}, n = 4$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$16\,700\left(1 + \frac{3}{100}\right)^4$$

$$A = \$16\,700 \times 1.256$$

$$A = \$18\,796$$

$$A = P + I$$

$$\$18\,796 = \$16\,700 + I$$

$$I = \$18\,796 - \$16\,700$$

$$I = \$2096$$

Page 13 questions

- c** \$20 000 is invested at 8% pa compounded every 6 months

$$P = \$20\,000, r = \frac{8}{2} = 4\% \text{ per 6 months, } n = 2$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$20\,000\left(1 + \frac{4}{100}\right)^2$$

$$A = \$20\,000 \times \$1.0816$$

$$A = \$21\,632$$

$$A = P + I$$

$$\$21\,632 = \$20\,000 + I$$

$$I = \$21\,632 - \$20\,000$$

$$I = \$1\,632$$

Page 14 questions

You invest your \$375 for 2 years. Use the compound interest formula to calculate how much money you will have (to the nearest cent) if the interest rate is

- a** 7.2% pa compounded annually

$$P = \$375, T = 2, r = 7.2$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$375\left(1 + \frac{7.2}{100}\right)^2$$

$$A = \$430.94$$

- b** 7.2% pa compounded monthly

$$P = \$375, T = 2 \times 12 = 24, r = \frac{7.2}{12} = 0.6\% \text{ monthly}$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$375\left(1 + \frac{0.6}{100}\right)^{24}$$

$$A = \$432.90$$

- c** Which is a better investment?

7.2% pa compounded monthly is a better investment since there is a higher return.

Page 17 questions

How much money do you need to invest if you're aiming to have \$13 500 after 5 years and

- a the interest rate is 2.4% pa compounded annually?

$$A = \$13\,500, r = 2.4, T = 5$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$\$13\,500 = P\left(1 + \frac{2.4}{100}\right)^5$$

$$P = \$11\,990.41$$

- b the interest rate is 2.4% pa compounded monthly?

$$A = \$13\,500, r = \frac{2.4}{12} = 0.2\% \text{ monthly}, T = 5 \times 12 = 60 \text{ months}$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$\$13\,500 = P\left(1 + \frac{0.2}{100}\right)^{60}$$

$$P = \$11\,974.86$$

Page 18 questions

Find the annual interest rate, to have \$5000 in 7 years if you invest \$2360 to two decimal places if the rate is

- a Compounded annually

$$A = \$5000, P = \$2360, T = 7$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$\$5000 = \$2360\left(1 + \frac{r}{100}\right)^7$$

$$1 + \frac{r}{100} = \sqrt[7]{\frac{\$5000}{\$2360}}$$

$$1 + \frac{r}{100} = 1.1132$$

$$\frac{r}{100} = 1.1132 - 1$$

$$\frac{r}{100} = 0.1132$$

$$r = 11.32\% \text{ pa}$$

- b Compounded monthly

$$n = 7 \times 12 = 84 \text{ months}$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$\$5000 = \$2360\left(1 + \frac{r}{100}\right)^{84}$$

$$\left(1 + \frac{r}{100}\right)^{84} = \frac{\$5000}{\$2360}$$

$$1 + \frac{r}{100} = \sqrt[84]{\frac{\$5000}{\$2360}}$$

$$1 + \frac{r}{100} = 1.00897$$

$$\frac{r}{100} = 1.00897 - 1$$

$$\frac{r}{100} = 0.00897$$

$$r = 0.897\% \text{ per month}$$

Page 19 questions

A principal investment P is invested for 10 years. Answer these questions correct to 2 decimal places

- a Use the simple interest formula to find the simple interest rate required for the investment to double.

$$A = P + I$$

$$\text{Let } A = 2P$$

$$2P = P + I$$

$$I = P$$

$$I = PRT$$

Substitute $I = P$ and $T = 10$ into $I = PRT$

$$P = P \times R \times 10$$

$$1 = R10$$

$$R = \frac{1}{10} = 0.1$$

$$R = \frac{r}{100}$$

$$0.1 = \frac{r}{100}$$

$$r = 10.00\% \text{ pa}$$

- b Use the compound interest formula to find the interest rate compounded annually required to double the principal.

$A = 2P$ since the final amount is going to be double the principal

$$A = P\left(1 + \frac{r}{100}\right)^n$$

Substitute $A = 2P$, $n = 10$ into $A = P\left(1 + \frac{r}{100}\right)^n$

$$2P = P\left(1 + \frac{r}{100}\right)^{10}$$

$$\left(1 + \frac{r}{100}\right)^{10} = \frac{2P}{P}$$

$$1 + \frac{r}{100} = {}^{10}\sqrt{2}$$

$$1 + \frac{r}{100} = 1.07$$

$$\frac{r}{100} = 1.07 - 1$$

$$r = 7.00\% \text{ pa compounded annually}$$

Page 20 questions

How long will it take to earn a total investment greater than \$60 000 if you invest \$17 000 at interest rate of

- a 18% pa compounded annually

$$A = \$60\,000, P = \$17\,000, r = 18\%$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$17\,000\left(1 + \frac{18}{100}\right)^n$$

if $n = 6$

$$A = \$17\,000\left(1 + \frac{18}{100}\right)^6$$

$$A = \$17\,000\left(\frac{118}{100}\right)^6$$

$$A = \$45\,982.42 \text{ (To the nearest cent)}$$

This is less than \$60 000

if $n = 7$

$$A = \$17\,000\left(1 + \frac{18}{100}\right)^7$$

$$A = \$17\,000\left(\frac{118}{100}\right)^7$$

$$A = \$54\,153.06 \text{ (To the nearest cent)}$$

This is less than \$60 000

if $n = 8$

$$A = \$17\,000\left(1 + \frac{18}{100}\right)^8$$

$$A = \$17\,000\left(\frac{118}{100}\right)^8$$

$$A = \$63\,900.61 \text{ (To the nearest cent)}$$

This is more than \$60 000

It will take 8 years for the investment to be greater than \$60 000

Page 20 questions

- b** 18% pa compounded monthly

$$r = \frac{18}{12} = 1.5\% \text{ monthly}$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$17\,000\left(1 + \frac{1.5}{100}\right)^n$$

Try $n = 80$

$$A = \$17\,000\left(1 + \frac{1.5}{100}\right)^{80}$$

$$A = \$55\,941.27 \text{ (To the nearest cent)}$$

This is less than \$60 000

if $n = 85$

$$A = \$17\,000\left(1 + \frac{1.5}{100}\right)^n$$

$$A = \$17\,000\left(1 + \frac{1.5}{100}\right)^{85}$$

$$A = \$60\,264.63 \text{ (To the nearest cent)}$$

This is just above \$60 000

It will take 85 months to earn more than \$60 000

- c** Which is quicker?

It would be quicker to invest in an interest rate that is compounded monthly in order to reach the \$60 000 faster.

Page 23 questions

Use the table converting annual reducible interest rates to equivalent annual flat rates to answer these questions.

- a Convert the 7% pa reducible interest over 6 years to a flat rate
3.79% pa
- b What is the equivalent annual flat rate for 9% pa over 3 years?
4.83%
- c What is the equivalent annual rate for 15% pa over 8 years?
9.03%
- d What is the equivalent annual flat rate to 1.25% per month over 5 years?
8.55%
- e What is the equivalent annual flat rate for 0.75% per month over 9 years?
5.14%

If to pay a loan off, you are required to pay a total of \$11 232 over 6 years then

- a If you are required to pay yearly, what is the annual payment?
 $\$11\,232 \div 6 = \1872 pa
- b If you are required to pay monthly, what is the monthly rate
 $T = 6 \text{ years} = 6 \times 12 \text{ months} = 72 \text{ months}$
 $\$11\,232 \div 72 = \156 per month
- c If you are required to pay weekly, what is your weekly payment?
 $T = 6 \text{ years} = 6 \times 52 \text{ weeks} = 312 \text{ weeks}$
 $\$11\,232 \div 312 = \36 per week

Page 24 questions

A bank loans Gary \$32 000. He must pay the loan off over the next 8 years at a rate of 11% pa compounded annually.

- a Do the time periods match the rate the interest is compounded?

Yes, since the interest rate is compounded annually and the time period specifies a year figure the time period and the period the interest rate is compounded do match.

- b Does the interest rate match the rate the interest rate is compounded?

Yes, the interest rate is a per annum figure and the interest rate is compounded annually then the interest rates match up.

- c Find the total amount Gary will have paid after he has paid off his loan.

$$P = \$32\,000, r = 11\%, n = 8$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$A = \$32\,000\left(1 + \frac{11}{100}\right)^8$$

$$A = \$73\,745.21$$

Page 25 questions

- d If Gary is required to pay this loan off monthly, how much will he have to pay each month?

$$A = \$73\,745.21$$

$$T = 8 \text{ years} = 8 \times 12 \text{ months} = 96 \text{ months}$$

$$A \div T = \$73\,745.21 \div 96 = \$768.18 \text{ per months}$$

- e If Gary is required to pay this loan off quarterly, how much will he have to pay each quarter?

$$A = \$73\,745.21$$

$$T = 8 \text{ years} = 8 \times 4 \text{ quarters} = 32 \text{ quarters}$$

$$A \div T = \$73\,745.21 \div 32 = \$2304.54 \text{ per months}$$

Page 26 questions

Let's say you borrow \$100 000 from a bank to put a deposit on a house. The bank charges you interest at 16.8% pa compounded monthly for the next 8 years.

- a Convert the annual interest rate compounded monthly to a monthly rate compounded monthly. Why is this necessary?

$$16.8 \div 12 = 1.4\% \text{ per month}$$

This is necessary since the formulas only work when units are consistent. For example, since the interest rate is compounded monthly it wouldn't make sense to use years as your time period since the interest rate is not compounded annually.

- b Find the total amount you would have paid the bank after 8 years.

$$A = \$100\,000, r = 1.4\%, n = 8 \times 12 = 96 \text{ per months}$$

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$$A = \$100\,000 \left(1 + \frac{1.4}{100} \right)^{96}$$

$$A = \$379\,877.60$$

Page 27 questions

- c How much profit did the bank make?

$$I = A - P$$

$$I = \$379\,877.60 - \$100\,000$$

$$I = \$279\,877.60$$

- d What is the value of each payment?

$$T = 8 \text{ years} = 8 \times 12 \text{ months} = 96 \text{ months}$$

$$A \div T = \$379\,877.60 \div 96 = \$3957.06$$

Page 28 questions

Ashley takes out a loan of \$3600. She is required to pay the loan back over 3 years at a reducible interest rate of 9%.

- a Find the equivalent annual flat interest rate from the table.

4.83%

- b Use the flat interest rate to find the total Ashley will have paid over the 3 years in paying off her loan.

$$P = \$3600, r = 4.83\%, T = 3$$

$$I = P \times \frac{r}{100} \times T$$

$$I = \$3600 \times \frac{4.83}{100} \times 3$$

$$I = \$521.64$$

- c If Ashley is required to make quarterly payments, then how much must she pay each quarter?

$$A = P + I$$

$$A = \$3600 + \$521.64$$

$$A = \$4121.64$$

$$T = 3 \text{ years} \times 4 \text{ quarters} = 12 \text{ quarters}$$

$$A \div T = \$4121.64 \div 12 = \$343.47 \text{ per quarter}$$

- d If Ashley is required to make monthly payments, then how much is Ashley required to pay each month.

$$T = 3 \text{ years} \times 12 \text{ months} = 36 \text{ months}$$

$$A \div T = \$4121.64 \div 36 = \$114.49 \text{ per months}$$

Page 29 questions

Find your monthly payments if you take out a loan of \$8500 at a reducible interest rate of 11% pa over 10 years.

Annual rate = 6.53% from the table

$$P = \$8500, r = 11\%, T = 10$$

$$I = P \times \frac{r}{100} \times T$$

$$I = \$8500 \times \frac{11}{100} \times 10$$

$$I = \$5550.50$$

$$A = I + P$$

$$A = \$5500.50 + \$8500$$

$$A = \$14\,050.50$$

$$T = 10 \times 12 = 120 \text{ months}$$

$$A \div T = \$14\,050.50 \div 120$$

$$A \div T = \$117.09 \text{ per month}$$

